

Quantum Gravitation

Problem Set 1

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Due 2026-09-06 23:59

Topic

Parameters of the Low-Energy Model

Submission

Submit one PDF. Show the derivation, identify every approximation, and include a short consistency check. The maximum file size is 25 MB.

Problems

1. Derive the first displayed relation used in Parameters of the Low-Energy Model. State all assumptions and conventions.

$$N_{SM} = N_g + N_{other} = 47 + 19 = 66$$

2. Calculate a nontrivial case using the second relation. Carry units and uncertainty where applicable.

$$p^\mu = (E/c, p)_{F01}$$

3. Verify the third relation in two limiting regimes and explain the physical meaning of the result.

$$L_{66} = \sum_b L_b$$

Show enough intermediate work that another student can reproduce the calculation.